# Utkarsh Pathak | DA24S011

**Indian Institute of Technology Madras**

Wadhwani School of Artificial Intelligence and Data Science

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## Education

| Program | Institution | %/CGPA | Year |
|---|---|---|---|
| MS Research* -ongoing (DSAI) | Indian Institute of Technology, Madras | **8.02/10** | 2025* |
| B.E. (CSE) | Chameli Devi School of Engineering (CDGI) | **8.33/10** | 2020 |
| XIIth Std. (PCM) | Aaditya Vidhya Vihar Hr. Sec. School | **85.3%** | 2016 |
| Xth Std. | Aaditya Vidhya Vihar Hr. Sec. School | **8.8/10** | 2014 |

## Research Activities

**1. Kinship in Speech: Leveraging Linguistic Relatedness for Zero-Shot TTS in Indian Languages**

*ACCEPTED at Conference :* ***INTERSPEECH 2025, Rotterdam*** — *Publication*

- Proposed a Zero-shot text-to-speech (TTS) framework for low-resource Indian languages by leveraging linguistic connections across language families. The method combines an extended shared phoneme set (called Common Label Set (CLS)) with phonotactic-aware text parsing and synthesizer selection, enabling intelligible and natural speech synthesis for unseen languages without parallel data.
- We showed that Zero-shot TTS is possible for languages with scripts and phonotactics from different language families (Ex: Sanskrit). We also demonstrated the effectiveness of dialect-based zero-shot synthesis for dialects across language families (Ex: Maharashtrian and Canara Konkani).
- This work is foundational as it enables speech synthesis across dialectal variations and unseen languages of the world.

## Work Experience

- **Project Officer in 'Speech Technologies in Indian Languages'** — **March 2023 - Ongoing**
  *Center for Industrial Consultancy & Sponsored Research, IIT Madras* — *Chennai*

- Funded by Ministry of Electronics and Information Technology (MEITY), the project aims to develop a robust and scalable system for **translating NPTEL educational videos** from English into regional languages, ensuring accuracy, accessibility, and inclusivity in education.
- Trained 22 official language TTS models from scratch using **Fastspeech2** (ESPNET toolkit) and **HiFi-GAN** vocoders to generate high-quality audio files and those models were deployed in Translingua annotation tool by NetPhenix startup, and on NPTEL servers. Also, **optimised** the models so that they can be used on CPUs.
- Developed a **T5-Transformer** based model for Hindi phoneme prediction, treating it as Seq2Seq problem, improving accuracy, enabling scalable **grapheme to phoneme** conversion for Indian languages.
- Helped develop a **robust pipeline for generating lip-sync videos**. The workflow included: processing video files with Automatic Speech Recognition (ASR) module to generate subtitles (SRT) files, translating subtitles using Machine Translation (MT) module, converting text to speech via TTS module, and synchronizing audio with video using lip-sync module. This end-to-end solution streamlined video production and enhanced accessibility of NPTEL videos.
- **Open-sourced** all of the lab's work on the official website and Github Repos.

- **Senior Systems Engineer in 'Concierge Project' for Microsoft D365 CRM** — **Jan 2021 - March 2023**
  *Infosys Limited* — *Pune*

- Understand the clients requirement by going through the specification and with inputs from Analysts. Help resolve **Bugs** that may arise in the product. Also, develop and review Artifacts(Code, Documentation, Tests) and make it ready for delivery.
- Work on **'Go Live'** activities and respond to production issue. Document and share own learnings from the project. Prepare KT Sessions and document for support.
- **Technologies** : .NET, C#, Microsoft Azure Cloud services, Angular, mySQL.

- **Full Stack Web development Intern** **Dec 2019 - Feb 2020**
  *Ypsilon IT Solutions* *Indore*

- Worked on a Web Application based on Travel booking using JAVA, MySQL Server and View on ANGULAR.

## Key Projects

- **Domain Term Discovery and Transcription correction for MT** **(Prediction)**
  *Team : 2* *(Project work)*

- Improved transcription accuracy for extempore-style NPTEL/Educational lectures by using a locally deployed LLM for post-editing.
- Focused on **preserving domain-specific terminology** often mishandled by machine translation systems.
- Curated a large dataset using GPT-3.5 Turbo to **train BERT models for automatic domain term identification**. Also, experimented with **few-shot learning** to identify and preserve key terms on-the-fly, significantly **reducing manual verification effort** and associated costs.

- **Transformer T5 model for Phoneme prediction** **(seq2seq + NLP)**
  *(Project work)*

- Developed a **T5-Transformer** based model for Hindi phoneme prediction, treating it as Seq2Seq problem, improving accuracy, enabling scalable **grapheme to phoneme conversion for Indian languages.**
- Trained it on large curated dataset generated using Unified Parser.

## Course Projects

- **CS6910: Fundamentals of Deep Learning** **Jan-May 2025**
  *(Prof. Mitesh Khapra)* *IIT Madras*

- Implemented the Neural Networks and Backpropogation from scratch and Optimization algorithms like SGD, Momentum, NAG, **Adam**, RMSProp, Adagrad.
- Developed a **CNN** based image classifier using a iNaturalist dataset and also did feature visualizations and also performed image classification by **Fine Tuning** different pre-trained models.
- Performed thorough analysis between **RNN, LSTM, GRU** for a **Machine Transliteration** on Dakshina Dataset. Improved the performance of vanilla models by adding **Attention Mechanism.**

- **CS5691: Pattern Recognition and Machine Learning** **Jan-May 2024**
  *(Prof. Arun Rajkumar)* *IIT Madras*

- Applied dimensionality reduction and unsupervised learning techniques through hands-on assignments and mini-projects. Gained practical experience in implementing **PCA and its kernel-based variants** for feature extraction, and developed clustering models using **GMMs optimized via the EM algorithm**.
- Implemented **Naive Bayes** from scratch to identify spam and non-spam emails. Implemented **Adaboost** algorithm.

- **CS6370: Natural Language Processing** **Jan-May 2025**
  *(Prof. Sutanu Chakraborti)* *IIT Madras*

- Covered foundational and advanced concepts in NLP including **language modeling, syntactic parsing, POS tagging, semantic analysis, and machine translation.**
- Gained hands-on experience through programming assignments and **implemented an Information retrieval system.**

## Relevant Course Work

- Pattern Recognition and Machine Learning (CS5691), Deep learning (CS6910), Natural Language Processing (CS6370), Linear Algebra, Random Probabilities and Optimisation(DA5000), Advanced Topics in Signal Processing (EE7130).

## Technical Skills

- **Programming Language**: C, C++, Java, Python
- **Libraries and tools**: Pytorch, HuggingFace, Sklearn, Numpy, Pandas, Android, Apache
- **Core concepts**: Operating systems, DBMS, Computer Networks
- **Web Development**: HTML, CSS, JavaScript